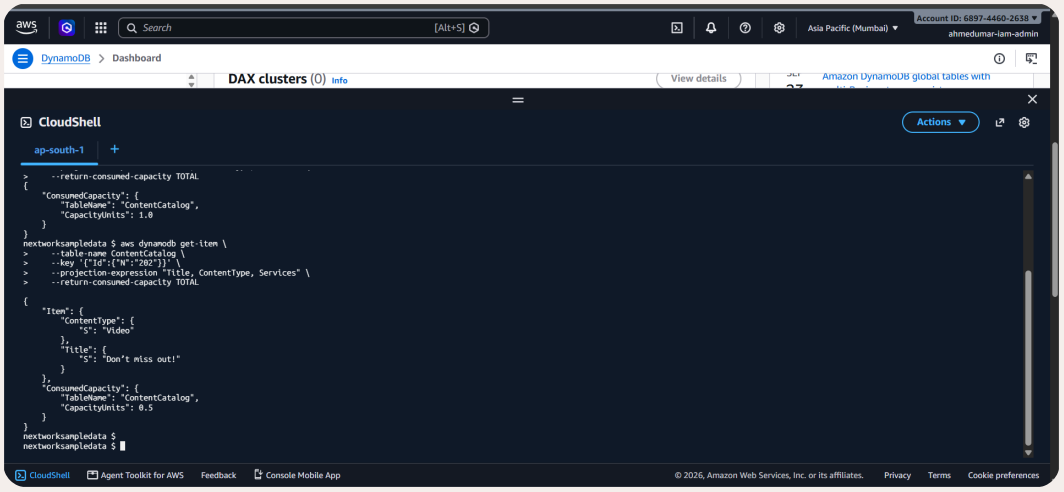
QUERY DATA WITH DYNAMODB

By Ahmed Umar Rehman | October 2026



Introducing Today's Project!

What is Amazon DynamoDB?

Amazon DynamoDB is a non-relational (NoSQL) database that stores data in tables, items, and attributes.
It is useful because it provides:
- Flexibility — each item can have its own set of attributes.
- Fast data access — partition keys help DynamoDB quickly find items.
- Scalability — it can handle large amounts of data and requests.
- Efficient queries — partition and sort keys help retrieve specific data

How I used Amazon DynamoDB in this project

In today's project, I used Amazon DynamoDB to create tables, load data into them using AWS CloudShell, query the data, and update related tables using transactions.

One thing I didn't expect in this project was...

One thing I didn't expect in this project is how important data modelling is for DynamoDB queries, especially the need to use the partition key when querying data.

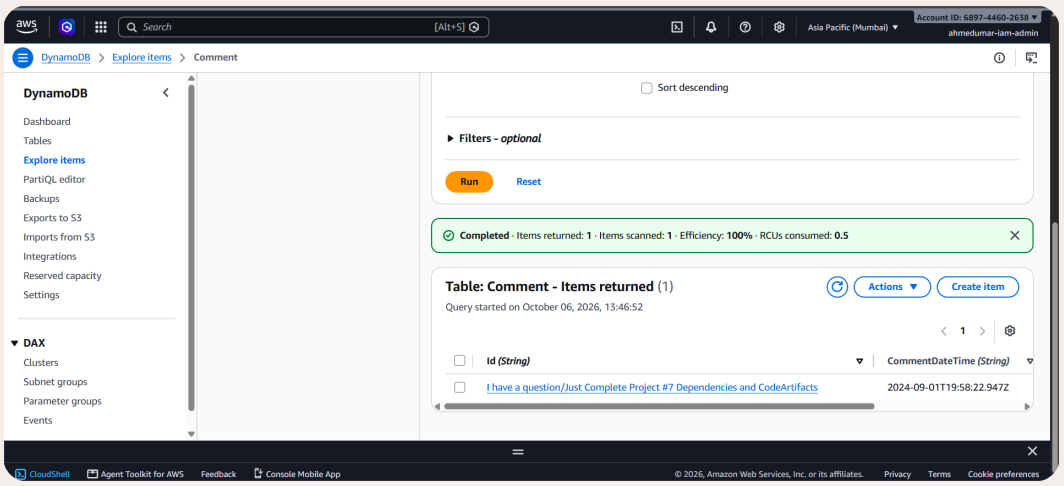
This project took me...

This project took me 40 mins

Querying DynamoDB Tables

A partition key is a key used to identify and locate items in a DynamoDB table.

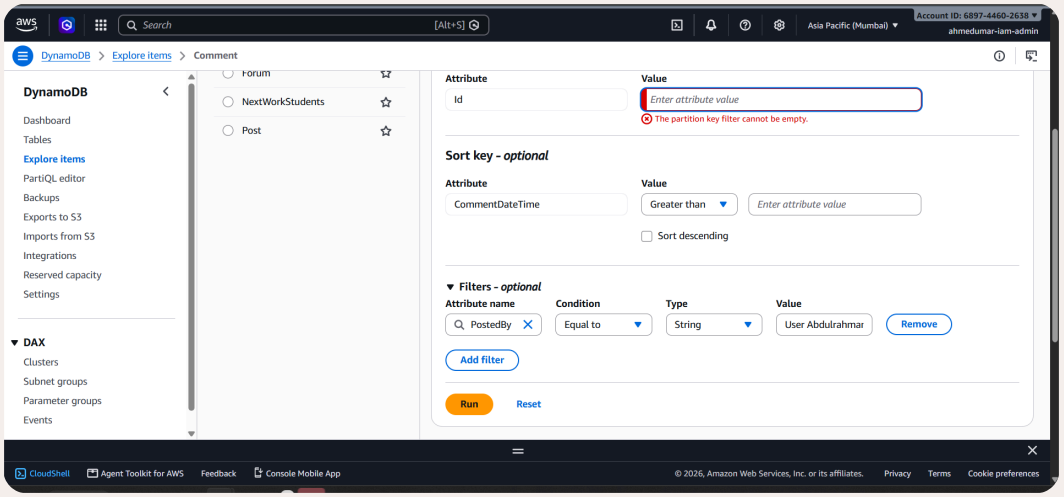
A sort key is an optional key used to organize and sort items with the same partition key, allowing you to find related data more precisely.



Limits of Using DynamoDB

I ran into an error when I queried for AbdurRehman. This was because I tried to query using an attribute that was not the table's partition key.

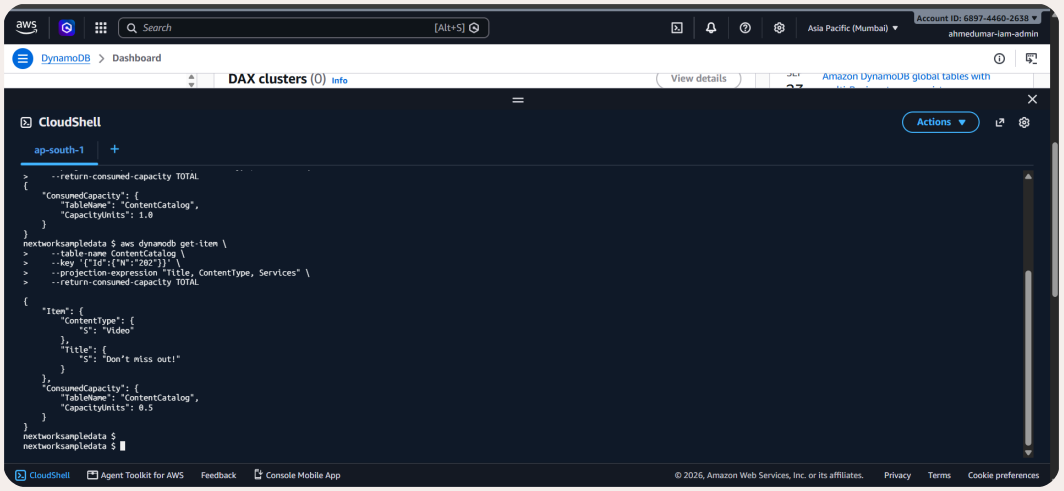
Insights we could extract from our Comment table include the comments associated with a specific post and the users who posted those comments.



Running Queries with CLI

This query will retrieve the item with Id = 101 from the ContentCatalog table and return only its Title, ContentType, and Services attributes, while using a strongly consistent read and showing the consumed capacity.

Query options I could add to my query are --consistent-read, --projection-expression, and --return-consumed-capacity to control read consistency, select specific attributes, and check the capacity consumed by the request.



Transactions

A transaction is a group of operations that must all succeed; if one operation fails, none of the changes are applied.

I ran a transaction using the `transact-write-items` AWS CLI command. This transaction did two things: added a new comment to the `Comment` table and increased the `Comments` count in the related `Forum` table by 1.

